

ARTERN, Germany

WMO: 104600

Lat: 51.3744N

Lon: 11.2919E

Elev: 166

StdP: 99.35

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.5	-10.8	-16.0	0.9	-13.0	-13.2	1.2	-10.5	14.1	5.6	12.6	4.9	1.9	40	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.3	19.3	27.5	18.5	25.8	17.7	20.2	27.2	19.2	25.7	18.4	24.3	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.0	22.6	16.8	12.2	21.5	16.0	11.6	21.0	58.5	27.3	55.3	25.9	52.6	24.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.4	DB	-14.3	33.6	5.9	1.9	-18.5	35.0	-21.9	36.1	-25.2	37.2	-29.5	38.5	(4)
(5)			WB	-14.6	22.2	5.8	1.2	-18.8	23.1	-22.1	23.8	-25.4	24.5	-29.5	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	0.3	-0.1	4.5	8.1	13.0	15.9	18.3	18.3	14.5	9.7	4.3	1.8
	DBStd	7.69	5.68	5.37	4.05	3.75	3.46	3.40	3.14	3.19	2.98	3.74	3.91	4.23
	HDD10.0	1331	300	284	172	79	12	1	0	0	2	50	174	256
	HDD18.3	3470	558	517	428	307	166	88	41	42	121	267	422	514
	CDD10.0	1003	0	1	3	23	106	177	257	256	136	42	2	1
	CDD18.3	102	0	0	0	0	2	15	40	40	5	1	0	0
	CDH23.3	1030	0	0	0	2	38	152	396	382	56	3	0	0
	CDH26.7	255	0	0	0	0	2	32	102	113	7	0	0	0
Wind	WSAvg	3.3	4.2	3.9	3.9	3.6	3.0	3.0	2.8	2.8	3.0	3.2	3.4	3.5
Precipitation	PrecAvg	574	38	34	41	36	60	57	71	55	51	39	47	46
	PrecMax	808	68	68	94	90	129	111	136	106	143	120	104	90
	PrecMin	355	6	8	8	7	4	23	19	19	15	7	2	13
	PrecStd	97	16	17	23	17	32	21	30	23	28	25	22	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	14.5	18.3	23.0	26.2	30.0	31.5	32.9	27.7	22.5	14.3	12.2
		MCWB	8.8	10.3	11.4	13.7	16.8	19.7	20.8	19.7	17.6	16.3	11.1	9.4
	2%	DB	9.5	10.9	14.2	19.6	24.3	27.2	29.3	29.5	24.9	19.3	12.2	10.6
		MCWB	7.4	7.9	9.5	12.3	15.9	18.7	19.6	19.0	16.8	14.7	10.1	8.5
	5%	DB	8.2	8.4	11.9	17.1	22.2	24.9	27.5	27.4	22.3	16.9	10.9	9.2
		MCWB	6.3	6.2	8.0	10.9	14.9	17.4	18.7	18.2	16.2	13.1	8.9	7.5
	10%	DB	7.0	6.2	10.2	14.9	20.2	22.6	25.6	25.2	20.0	15.2	9.6	7.7
		MCWB	5.3	4.5	7.3	9.8	14.0	16.5	17.9	17.5	15.3	12.1	7.9	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.3	11.7	14.1	18.3	21.4	21.9	21.1	19.4	17.4	11.7	9.8
		MCDB	11.3	14.1	16.2	21.1	24.4	27.7	29.8	31.3	24.6	21.9	13.2	11.7
	2%	WB	7.6	8.1	9.9	12.7	16.7	19.3	20.4	19.8	17.9	15.1	10.2	8.7
		MCDB	9.3	10.4	14.4	18.7	22.8	25.2	27.6	27.0	22.8	18.5	11.9	10.5
	5%	WB	6.5	6.1	8.6	11.3	15.5	18.2	19.3	18.9	16.8	13.6	9.2	7.7
		MCDB	8.1	8.1	11.4	16.2	21.1	23.7	26.0	25.4	21.3	16.4	10.8	9.3
	10%	WB	5.3	4.6	7.4	10.1	14.3	16.9	18.4	18.1	15.6	12.4	8.0	6.2
		MCDB	7.0	6.1	10.0	14.5	19.4	21.3	24.2	24.1	19.7	14.7	9.4	7.5
Mean Daily Temperature Range	5% DB	MDBR	4.7	6.0	7.4	9.1	10.3	9.6	10.8	10.8	9.4	8.0	5.4	4.1
		MCDBR	5.0	8.2	11.3	13.0	13.9	13.5	14.5	14.8	13.3	11.0	6.4	5.4
		MCWBR	4.3	6.0	7.3	7.1	7.3	6.6	6.5	6.4	6.9	6.9	5.0	4.5
	5% WB	MCDBR	4.9	7.7	10.5	11.8	12.6	12.4	13.3	12.5	11.9	10.5	6.4	5.3
		MCWBR	4.4	5.9	7.1	6.7	6.9	6.6	6.4	5.8	6.6	6.9	5.1	4.5
Clear-Sky Solar Irradiance	taub	0.308	0.330	0.370	0.408	0.412	0.408	0.424	0.417	0.381	0.354	0.330	0.303	
	taud	2.406	2.371	2.278	2.215	2.236	2.271	2.231	2.271	2.358	2.424	2.429	2.399	
	Ebn at Noon	712	792	820	828	843	848	825	810	801	753	675	653	
	Edh at Noon	60	81	107	128	131	128	131	120	98	77	59	52	
All-Sky Solar Radiation	RadAvg	0.73	1.43	2.51	4.01	4.89	5.29	4.97	4.26	3.08	1.79	0.83	0.55	
	RadStd	0.10	0.20	0.31	0.49	0.57	0.44	0.65	0.41	0.40	0.20	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A

Nomenclature: See separate page